

Four Ways to Adjust: The Tutoring Example

Experimentation, A/B Testing and Causal Inference (SDA-DSC-213), SDAIA Academy. Handout for Day 3, Hour 2 (slides 94 to 100). All five Day 3 handouts use the same story: a tutoring programme at one school.

Step 1. The story and the data

Same eight students as the DAG handout. X is prior grade, T is tutoring, Y is the exam score. The DAG said: adjust for X and nothing else. This handout computes the same effect four ways (slide 95).

Student	X	T	Y
A	Low	1	60
B	Low	1	62
C	Low	1	64
D	Low	0	52
E	High	1	88
F	High	0	78
G	High	0	80
H	High	0	82

Estimand **estimand: τ , the ATE**: the average effect over all eight students. Identifying assumption: within a level of X, who got tutoring is as good as random (unconfoundedness given X), and every student had a real chance of either arm (overlap).

Step 2. Outcome regression, the g-formula

Model the outcome given T and X, then predict both potential outcomes for every student and average the difference. With one binary X the model is just the four cell means.

$$E[Y \mid T=1, \text{Low}] = 62 \quad E[Y \mid T=0, \text{Low}] = 52 \quad E[Y \mid T=1, \text{High}] = 88 \quad E[Y \mid T=0, \text{High}] = 80$$

$$\hat{\tau}_g = (1/N) \sum_i [\hat{Y}_i(1) - \hat{Y}_i(0)] = (4 \times (62-52) + 4 \times (88-80)) / 8 = (40 + 32) / 8 = +9$$

Fails if the outcome model is wrong (for example, a straight line through a curved relationship). Here the model is saturated, so it cannot be wrong.

Step 3. The propensity score

$e(x) = P(T = 1 | X)$: the chance of being tutored given the prior grade (slide 96). Estimate it from the data:

$$e(\text{Low}) = 3 \text{ tutored} / 4 \text{ students} = 0.75 \quad e(\text{High}) = 1 / 4 = 0.25$$

Balancing property: within a level of $e(x)$, tutored and untutored students have the same X . Judge the score by the balance it produces, not by how well it predicts T .

Step 4. Inverse-probability weighting

Tutored students get weight $1/e(x)$; untutored get $1/(1 - e(x))$. Rare-in-arm students count for more, which rebuilds a pseudo-population where T no longer depends on X (slide 97).

Student	X	T	Y	Weight	Weight $\times$ Y
A	Low	1	60	$1/0.75 = 1.333$	80.0
B	Low	1	62	1.333	82.7
C	Low	1	64	1.333	85.3
E	High	1	88	$1/0.25 = 4$	352.0
D	Low	0	52	$1/(1-0.75) = 4$	208.0
F	High	0	78	$1/(1-0.25) = 1.333$	104.0
G	High	0	80	1.333	106.7
H	High	0	82	1.333	109.3

$$\text{weighted mean}(T=1) = (80 + 82.7 + 85.3 + 352) / (1.333 \times 3 + 4) = 600 / 8 = 75$$

$$\text{weighted mean}(T=0) = (208 + 104 + 106.7 + 109.3) / (4 + 1.333 \times 3) = 528 / 8 = 66$$

$$\tau_{IPW}^{\hat{}} = 75 - 66 = +9$$

Balance check: before weighting, 75 percent of the tutored are Low versus 25 percent of the untutored. After weighting, both arms are 50 percent Low (weights $1.333 \times 3 = 4$ of 8 in each arm). The weighted world looks like an experiment.

The danger: $e(x)$ near 0 or 1 gives huge weights and a noisy estimate. Trim to the region where both arms exist, or use stabilised weights. Here the largest weight is 4, which is fine.

Step 5. Matching

Pair each tutored student with an untutored student who has the same X , and average the within-pair differences. With the Low controls being one student (D), the three Low

tutored students all match to D.

$$A-D = 60-52 = 8 \quad B-D = 62-52 = 10 \quad C-D = 64-52 = 12 \quad E - \text{mean}(F,G,H) = 88 - 80 = 8$$

$$\tau_{\text{match}}^{\wedge} = (8 + 10 + 12 + 8) / 4 = +9.5$$

This is +9.5, not +9, because matching every treated student estimates the effect on the treated **estimand: ATT**: three of the four tutored students are Low, where the effect is 10. Matching changes the estimand; weighting and the g-formula target the ATE unless told otherwise.

Step 6. Doubly robust (AIPW)

Combine both models so that the estimate is right if **either** the outcome model or the propensity model is right (slide 98). For each student, start from the outcome-model prediction and add a weighted correction from the residual:

$$\tau_{\text{AIPW}}^{\wedge} = (1/N) \sum_i [\hat{Y}_i(1) - \hat{Y}_i(0) + T_i(Y_i - \hat{Y}_i(1))/e_i - (1-T_i)(Y_i - \hat{Y}_i(0))/(1-e_i)]$$

Here both models are exactly right, so every residual $Y_i - \hat{Y}_i$ is the within-cell deviation and the corrections cancel within each cell: $\hat{\tau}_{\text{AIPW}} = +9$. In Lab 3 the models are fitted with machine learning on two folds (cross-fitting), which keeps the interval honest.

Step 7. The four answers side by side

Method	Estimate	Targets	Needs	Fails when
Naive difference	-4.5	nothing valid	no confounding	X differs between arms
g-formula	+9	ATE	correct outcome model	outcome model misspecified
IPW	+9	ATE	correct propensity model, overlap	extreme weights
Matching	+9.5	ATT	good matches, overlap	poor matches, discarded units
AIPW	+9	ATE	either model correct	both models wrong

Three things to remember

1. All four adjust for the same X chosen from the DAG; they differ in what they model and how they fail.

2. Check overlap and balance (standardized mean difference below 0.1) before reading any estimate.
3. Prefer the doubly robust estimator with cross-fitting, report an interval, then follow with a sensitivity analysis.